

Moffat Bay Marina

Website Design Document



Introduction

This document will provide technical implementation details for a website that is to be created for Moffat Bay Marina. Within this document you will find a Terminology section for all stakeholders to reference.

Purpose

Moffat Bay Marina currently offers long-term slip reservations that can be booked in person, over the phone and via email. Their office manager tracks reservations with a spreadsheet that they manually update. Moffat Bay Marina does not currently have a website, so customers must call or email to request information regarding the size of slips and reservation dates available. This process requires careful coordination and manual data entry, which can be time-consuming.

Moffat Bay Marina would like a website that can provide details about the marina and slip availability to any visitors. They would like their website to manage reservations and user accounts for their customers. It is important to note that, while a customer account needs to be created in order to utilize the online booking system, the website itself should be viewable to any visitors regardless of customer account creation. Additionally, the reservation system will not process payments for customers and will not require payment to confirm a reservation.

Terminology

Marina/Business terms

Slip - a designated area in a marina where a boat can "park"

Harbormaster - a person who oversees harbor or marina traffic and manages anchorages

VHF - stands for "Very High Frequency" and is a bandwidth designation commonly used by marine radios

Long-Term Reservation - reservations that last for a season (several months) to an entire year, as opposed to short-term reservations, which are measured in days, weeks or months

Technical/System terms

Authentication - "in web security, it is the process of verifying the claimed identity of some entity, such as a user. This then makes it possible to decide whether to grant the user the access that they are requesting, such as being signed into a particular account." (MDN Web Docs, n.d.).

User - in most cases, this will refer to the customer, but can refer to any person who is using a system (in this case, the website)

User account - customer account

CustomerID - a unique identifier for each customer to be stored in the Moffat Bay Marina database

ReservationID - a unique identifier for each reservation made to be stored in the Moffat Bay Marina database

Session - a window of time in which a user accesses a website

Landing Page - the main page of a website through which a user can access other sections of the website. The landing page will typically contain basic information about the business and navigation elements, like a menu

Form / Form Field - in HTML, a form contains elements, like fields, that can receive user input (ex. Email field) or user interactions (ex. Clicking the "submit" button after filling out an email field)

Database - a system that collects and stores organized data. Databases make searching and structuring information easier and more efficient.

Open Source - publicly available code/designs

RDBMS - stands for Relational Database Management System

MySQL - an open-source RDBMS that is commonly used to manage databases for companies

Regular Expression (Regex) - a pattern-matching mechanism that can be used to check whether text conforms to a specific format. For example, a phone number is 10 digits and is typically formatted XXX-XXX-XXXX

ERD - stands for "Entity-Relationship Diagram"; this is a visual flowchart typically used in database design that illustrates how entities (ex. Customers, slip reservations) are related within a system

HTML - stands for "HyperText Markup Language"; this is the standard language that is used to create website content

CSS - stands for "Cascading Style Sheets"; used for the visual organization of HTML

JSP - stands for "JavaServer Pages"; a technology that is used to create web pages that can display changing information from a Java-based web application. For example, if a customer visits the Moffat Bay Marina website and logs into their account, the banner on the webpage might display their name in the upper right corner to indicate they are logged in

PII - stands for "Personally Identifiable Information"; this can be any data that can be used to identify a specific person. For example, a person's social security number would be considered PII

User Personas

User 1: Angela

Gender: Female

Age: 42

Occupation: Elementary Teacher

Education: College/Undergraduate

Marital Status: Married

Location: Local, lives in town year-round



Background

Angela and her family have had a 40 ft boat for six years now, and they book a long-term slip every season. She's got a full-time job and two kids with their own schedules, so she doesn't have time to sit on the phone dealing with reservations. She just wants something quick that works.

Angela's Needs

- An option to see what slips are open without calling the business
- Some kind of confirmation she can send to her husband once it's booked
- An account that remembers her info so she's not typing the same stuff in every season

Angela's User Stories

1. As a returning customer, I want to check slip availability online so I can plan my season without calling the office.
 2. As a busy parent, I want my account and boat info saved after I register so I don't have to re-enter them every time.
 3. As a slip holder, I want confirmation that I can save or forward the reservation details to share with my family.
 4. As a returning customer, I want to see my past reservations so I know which slip I had last season and can request it again.
 5. As a slip holder, I want to get a reminder before my reservation period ends so I'm not caught off guard when it's time to renew.
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User 2: Jesse

Gender: Male

Age: 47

Occupation: Harbormaster

Education: Community College/Vocational School

Marital Status: Married

Location: Local, lives in town year-round



Background

This is Jesse, he is a staff member at the Moffat Bay Marina. Jesse's goal is to keep long-term slip reservations organized and make sure customers are assigned to slips that fit their boats.

Jesse's Needs

- Accurate reservations
- Slip availability
- Efficient access to registered customer information
- An easy way to confirm reservations.

Jesse's User Stories

1. As a marina staff member, I want to view current slip availability so that I can know which long-term slips are still available.
2. As a marina staff member, I want customers to be matched with the right slip size based on the length of their boat.
3. As a marina staff member, I want to see confirmed reservations so that I can track available slips and avoid double bookings.

User 3: Walt

Gender: Male

Age: 68

Occupation: Retired Electrician

Education: High School

Marital Status: Widowed

Location: Winter Visitor



Background

Walt is a 68-year-old retired electrician and widower who escapes the snow each winter at Moffatt Bay Marina Lodge. From November through February he stays at the lodge for three to four months, keeps his boat in a dock slip, and spends most days out fishing. Since losing his wife, Walt values reliability, simplicity, and a sense of community.

Walt's Needs

- A reserved slip for his entire 3-4 month winter stay
- Easy, bundled booking that happens his lodge room with his boat slip
- Peace of mind that his boat is secure between fishing trips

Walt's User Stories

1. As a winter dock guest, I want to reserve a dock slip for a full 3–4-month block in one transaction, so I don't have to re-book or risk losing my spot month to month.
2. As a guest who is older, I would prefer to select a dock slip that is closer to the shore, so I don't have to haul my equipment as far when I take it to and from the lodge.
3. As a long-term slip holder, I would like clear, upfront pricing for a multi-month stay, including any seasonal or long-term discounts, so that I can budget for the winter without surprise fees.
4. As a boat owner staying through long periods of time, I want information on what happens during inclement weather, such as procedures, safety, and insurance information.
5. As a boat owner and slip holder, I would like the option to have marina staff check on my boat for me if I'm going to be gone for extended periods of time.

Work Estimations

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